Economic Downturn, Turnout, and Incumbent Support

Final Report – STA 702 Statistical Analysis Project

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Part 1 – Data Science Memo

Executive Summary

This project examines how **local economic downturns** relate to (1) **voter turnout** and (2) **support for the incumbent presidential party** in U.S. counties.

Using a county-year panel from **2004–2022** (turnout) and **2000–2020** (presidential elections), we link voting outcomes to county-level unemployment from federal data sources.

Our main findings are:

- **Turnout (RQ1).**
  Within a county, higher unemployment is associated with a **very small increase** in turnout (about **+0.3 percentage points** for a 2-point rise in unemployment). Cross-sectionally, high-unemployment counties tend to have lower turnout, but once we control for time-invariant county traits, the effect becomes small and positive. Structural factors such as election type (presidential vs midterm) matter far more than economic conditions.
- **Incumbent support (RQ2).**
  In contrast, unemployment has a **substantive and robust negative effect** on the probability that the incumbent presidential party wins a county. A one-percentage-point increase in the local unemployment rate reduces the odds of an incumbent victory by roughly **9–10%** in our fixed-effects logistic model.
- **Heterogeneous effects.**
  Economic voting is **strongest in rural counties** and **weakest in urban areas**. In recent elections, especially 2016 and 2020, some deeply distressed rural counties continued to support Republican incumbents despite high unemployment, reflecting a broader partisan realignment that partly offsets classical economic voting.

Overall, economic downturns only slightly change **who participates**, but they **meaningfully shift which party those voters support**, generally punishing incumbents in places where local labor markets have worsened.

Background and Motivation

Classic theories of political participation (Almond & Verba; Berelson et al.) view citizens as distributed along a **participation pyramid**: a small group of highly engaged "participants," a middle tier of “subjects” who vote in major elections, and a broad “parochial” or **silent majority** that rarely engages in politics. Economic crises are thought to activate parts of this silent majority and shift behavior among marginal voters.

At the same time, **retrospective economic voting** theory predicts that voters hold incumbents accountable for economic performance: rewarding them during prosperity and punishing them during downturns. Most empirical work focuses on national aggregates; our goal is to test whether **local labor-market conditions** at the county level shape both **turnout** and **incumbent vote share**.

We therefore ask two complementary questions:

- **Research Question 1 (RQ1 – Participation)**
  How does local unemployment relate to voter turnout in U.S. elections at the county level
- **Research Question 2 (RQ2 – Vote Choice)**
  Does local economic downturn affect the probability that a county supports the incumbent presidential party?

Data and Measures (High Level)

We construct a county-year panel by merging:

- **Voting outcomes** from the ICPSR *National Neighborhood Data Archive* (NaNDA), with turnout and two-party presidential results for U.S. counties from 2004–2022.
- **Unemployment rates** from the Bureau of Labor Statistics' *Local Area Unemployment Statistics* (LAUS).
- For RQ1 only, **Citizen Voting Age Population (CVAP)** to compute turnout rates and classify counties as **urban / suburban / rural** based on their voting-age population size.

Key variables:

- **Turnout outcome (RQ1)** – `VOTER_TURNOUT_PCT`: total ballots cast divided by CVAP, bounded between 0 and 1.
- **Incumbent outcome (RQ2)** – `incumbent_win`: indicator = 1 if the county is carried by the party of the sitting president in that election year, 0 otherwise.
- **Economic condition** – `unemployment_rate`: annual county unemployment rate (%). In RQ1 we also form **unemployment tiers** using z-scores (very low, low, medium, high, very high).
- **Structure / controls**
  - County fixed effects (`FIPS`) to absorb time-invariant traits (baseline partisanship, demographics, institutions).
  - Year fixed effects (`YEAR`) to absorb national shocks (wars, recessions, the pandemic).
  - County “type”: **urban / suburban / rural** for heterogeneity analysis.

The final merged dataset contains around **30,000 county-year observations** for turnout (2004–2022) and **six presidential elections** (2000, 2004, 2008, 2012, 2016, 2020) for incumbent support.

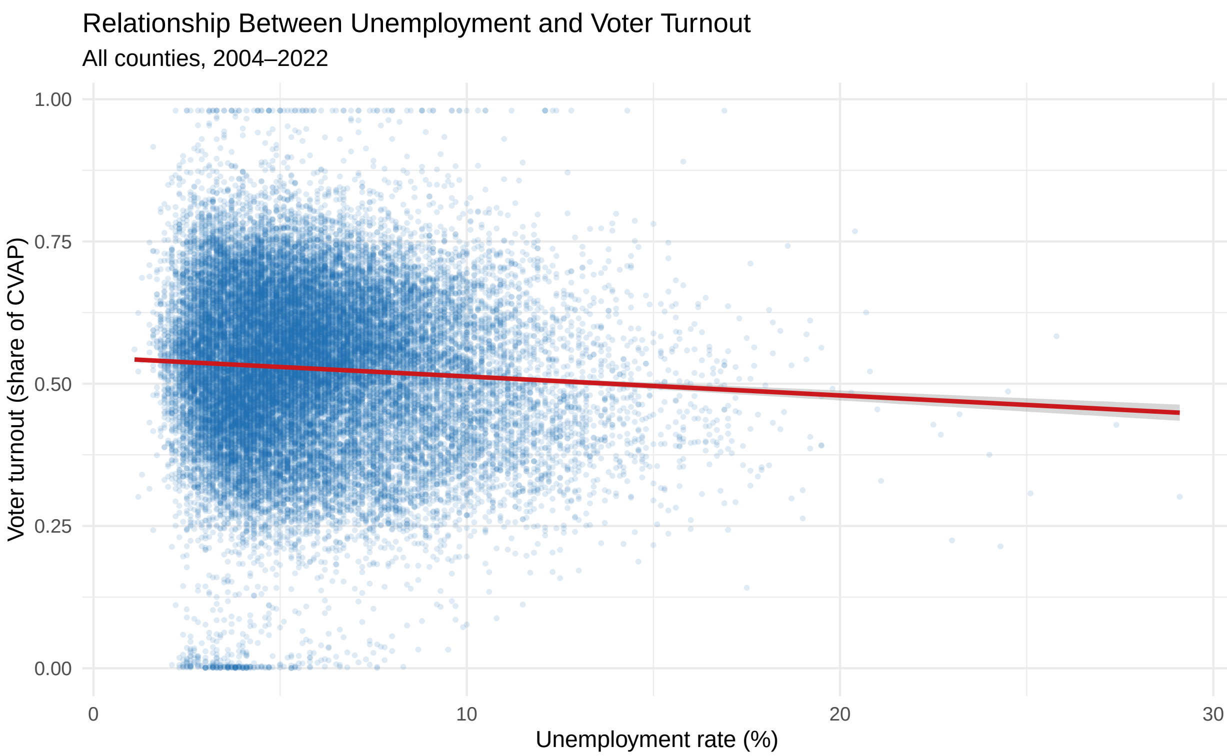
Key Results – Research Question 1 (Turnout)

Descriptive Patterns

Across all counties and years:

- Average turnout is around **50–55%** of the citizen voting-age population.
- Presidential elections have substantially higher turnout than midterms.
- High-unemployment counties tend to have slightly lower turnout in raw correlations.

We visualize this in a simple scatterplot with a linear fit:



Regression Evidence

For turnout, we estimate:

1. A **simple OLS** regressing turnout on unemployment.
2. A model **adding election type** (presidential vs midterm).
3. A model with **interaction** between unemployment and county type.
4. A **two-way fixed effects** model with county and year dummies.
5. A **fixed effects model with interaction** between unemployment and county type.

In the **simple bivariate model**, a 1 percentage point increase in unemployment is associated with a small **negative** change in turnout (around –0.3 percentage points), and unemployment explains less than 1% of the variation.

In the **county + year fixed effects model**, the sign flips: a 1-point increase in unemployment is associated with roughly **+0.16 percentage points** higher turnout within the same county over time. The magnitude is statistically detectable but substantively small.

A recession-level increase in unemployment of 2 percentage points (e.g., from 5% to 7%) is associated with only about **+0.3 percentage points** higher turnout in that county.

Interpretation for Stakeholders

- **Economic stress does not strongly demobilize voters.**
  Within a given county, higher unemployment is associated with a very small increase in turnout. This suggests that economic downturns may slightly activate otherwise marginal voters rather than discouraging participation.
- **Context dominates.**
  Whether an election is a **presidential or midterm** matters far more than local unemployment conditions. Institutional features (election timing, salience) and long-run county traits drive most of the variation in who turns out.
- **Heterogeneous responses.**
  Rural counties show somewhat stronger responsiveness of turnout to unemployment than urban ones, consistent with the idea that economically vulnerable communities may be more mobilized by deteriorating local conditions.

Key Results – Research Question 2 (Incumbent Support)

Constructing Incumbent Support

For each presidential election year (2000–2020), we:

1. Identify the **incumbent party** (Democratic or Republican).
2. Determine which party received more votes in each county.
3. Define `incumbent_win = 1` if the incumbent party carried that county, 0 otherwise.

Fixed Effects Logistic Model

We estimate:

$$\text{logit}\Big(\Pr(\text{incumbent\_win}_{c,y} = 1)\Big) = \beta \cdot \text{unemployment\_rate}_{c,y} + \alpha_c + \gamma_y + \varepsilon_{c,y}$$

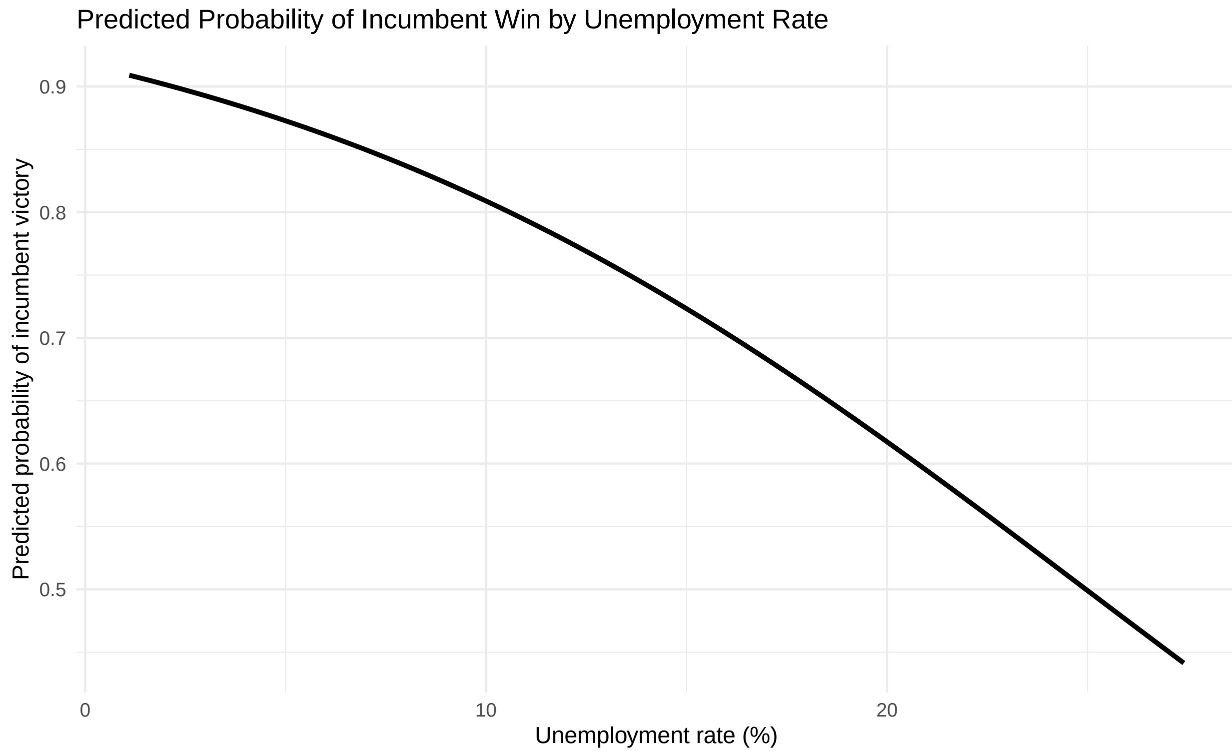
$$\alpha_c : \text{county fixed effects}$$

$$\gamma_y : \text{year fixed effects}$$

The estimated coefficient on `unemployment_rate` is **negative, statistically significant, and substantively meaningful** (about -0.096).

On the **log-odds scale**, a 1-point increase in unemployment reduces log-odds of an incumbent win by roughly 0.096. Converted to **odds**, this implies about a **9–10% reduction** in the odds of an incumbent victory for each additional percentage point of unemployment, holding county and year fixed.

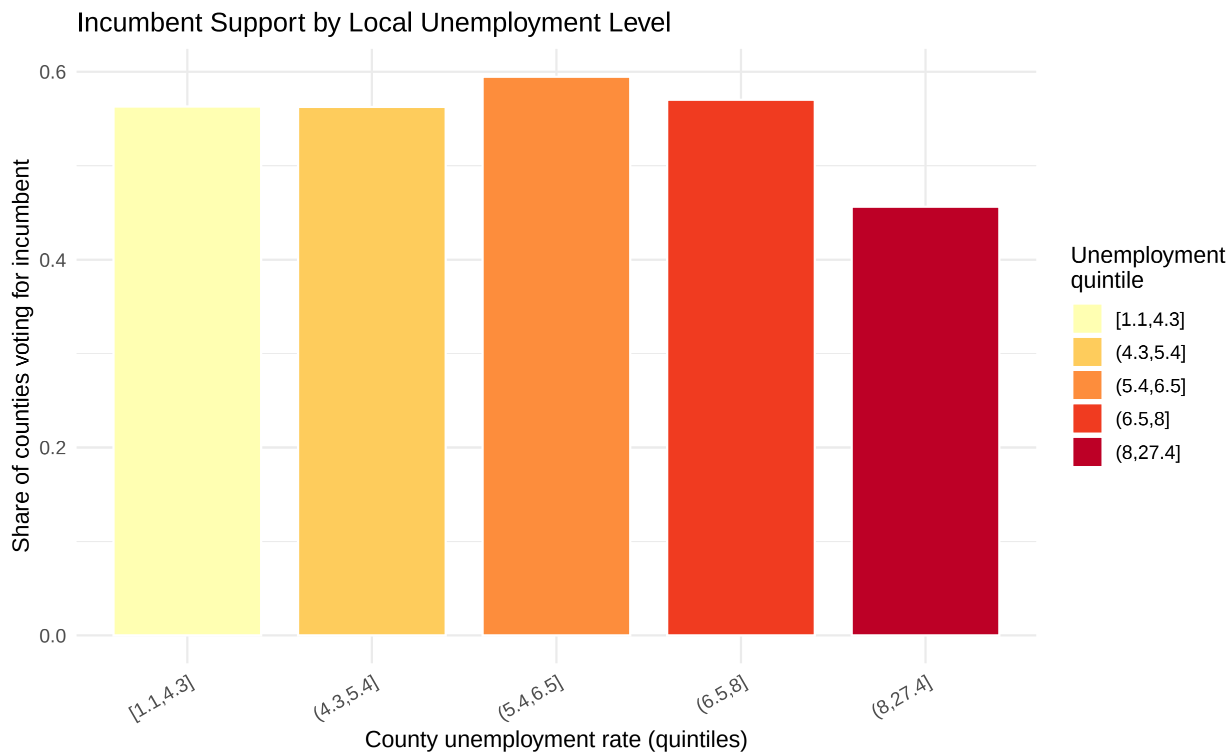
Model-Based Visualization



The curve slopes downward: counties with unemployment around **3–5%** have predicted incumbent support above **0.85**, while those above **15–20%** drop toward **0.45–0.50**.

Heterogeneous Patterns

We can summarize how incumbent support varies across the unemployment distribution:



Incumbent support is relatively stable across low and moderate unemployment levels but **drops sharply in the highest unemployment quintiles**, especially in rural counties.

Interpretation for Stakeholders

- **Economic voting is clearly present.**
  Local economic downturns significantly reduce the probability that a county votes for the president's party. Even modest increases in unemployment meaningfully tilt close counties away from incumbents.
- **Effect is stronger than on turnout.**
  While unemployment only slightly raises turnout, it **more strongly shifts vote choice** among those who participate. Voters seem to respond to local distress primarily by changing **who** they vote for rather than **whether** they vote.
- **Rural vulnerability and realignment.**
  Rural counties exhibit the largest substantive swings in incumbent support as unemployment rises, yet in the most recent elections some economically distressed rural counties remained strongly pro-incumbent Republican. This highlights the interaction between economic conditions and evolving partisan identities.

Policy Implications and Recommendations

From the perspective of a national campaign or policymaker:

1. **Economic management and communication remain central.**
   Local unemployment meaningfully harms incumbents, especially in competitive or economi-

cally vulnerable counties. Investments that stabilize local labor markets—or credible communication about recovery efforts—may have real electoral returns.

2. **Turnout strategies should focus on structural barriers.**
   Because unemployment has only a modest impact on turnout, reforms to **registration, early voting, mail ballots, and polling access** are likely more effective levers for changing participation than purely economic messaging.

3. **Targeting by county type.**
   - **Rural counties**: high sensitivity of incumbent support to economic distress suggests that visible local investment, infrastructure, and employment programs could mitigate electoral punishment.
   - **Urban counties**: economic voting is weaker; issues like policing, housing affordability, and social services may loom larger.

4. **Monitoring realignment.**
   The breakdown of classic economic voting in some rural areas after 2012 indicates that **partisan identity can override local economic pain**. Campaigns should not assume that improving economic conditions alone will reverse entrenched partisan trends.

Limitations and Future Work (High Level)

- **Measurement of economic conditions.**
  Unemployment is only one dimension of local distress. Future work could incorporate **wages, poverty, housing costs, plant closures**, or industry mix.
- **County-level aggregation.**
  Our analysis cannot recover individual-level mechanisms (e.g., whether unemployed individuals are more likely to switch parties). Micro-data would be needed for that.
- **Causality.**
  Fixed effects reduce bias from time-invariant traits and national shocks but cannot fully address **reverse causality** or omitted time-varying factors (e.g., local policy responses, campaign spending).
- **Limited number of presidential elections.**
  RQ2 relies on only six presidential cycles, so results may be sensitive to a few idiosyncratic elections.

Despite these caveats, the broad qualitative picture is robust across multiple model specifications and descriptive checks.

Part 2 – Technical Appendix

This appendix documents the main data construction steps, model specifications, and additional tables and figures for both research questions.

A. Data Construction and Cleaning

A.1 Turnout Dataset (RQ1)

A.2 Incumbent Support Dataset (RQ2)

B. Models and Detailed Results

B.1 Turnout Models (RQ1)

Effect of Unemployment on Voter Turnout - Full Model Set

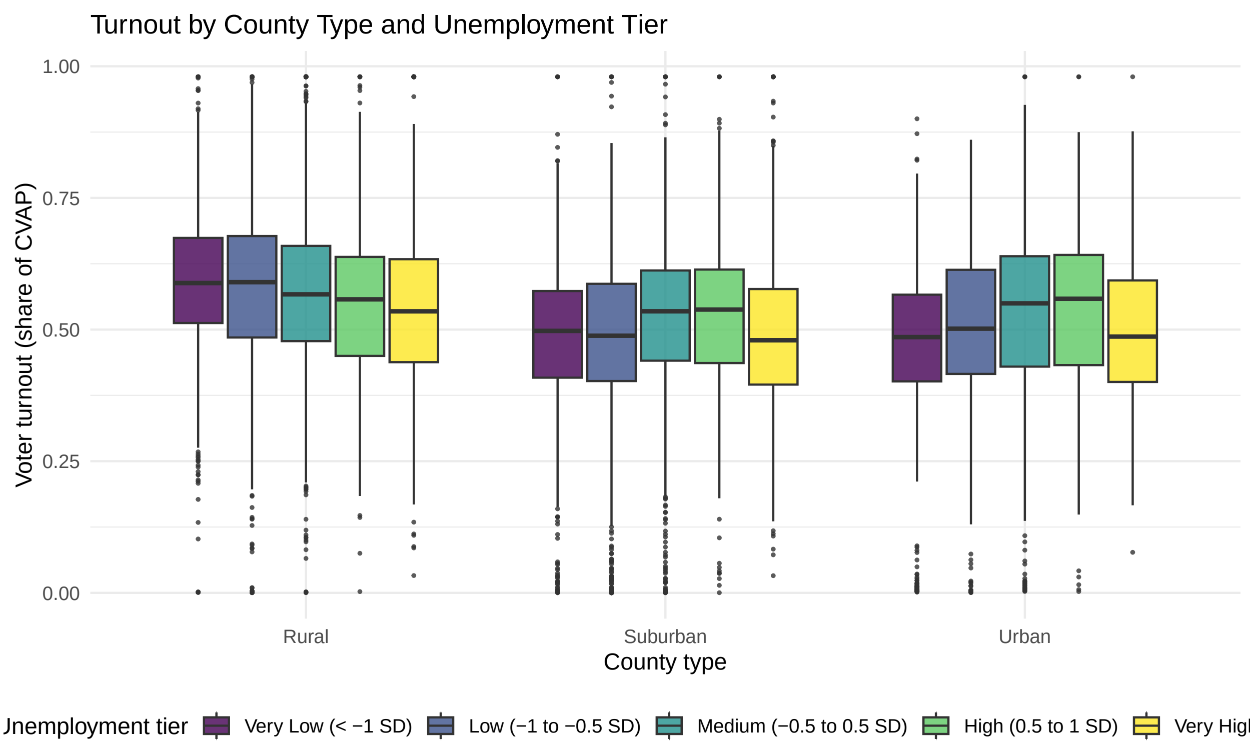
	Dependent variable:				
	Turnout				
	OLS	OLS + Election	OLS + Interaction	felm FE	felm FE + Int
	(1)	(2)	(3)	(4)	(5)
Unemployment rate	-0.0033***	-0.0074***	-0.0093***	0.0016***	0.0004
	(0.0003)	(0.0003)	(0.0004)	(0.0004)	(0.0006)
Presidential election			-0.0862***		-0.0362***
			(0.0037)		(0.0071)
County: Suburban			-0.0693***		-0.0477***
			(0.0043)		(0.0095)
County: Urban		0.1679***	0.1671***		
		(0.0014)	(0.0013)		
Unemp × Suburban			0.0051***		0.0015***
			(0.0006)		(0.0006)
Unemp × Urban			0.0037***		0.0020***
			(0.0007)		(0.0006)
Constant	0.5462***	0.4863***	0.5342***		
	(0.0020)	(0.0017)	(0.0027)		
Observations	29,769	29,769	29,769	29,769	29,769
R2	0.0038	0.3405	0.3715	0.7411	0.7415
Adjusted R2	0.0038	0.3405	0.3713	0.7109	0.7112
Note:					*p<0.1; **p<0.05; ***p<0.01

B.2 Incumbent Support Model (RQ2)

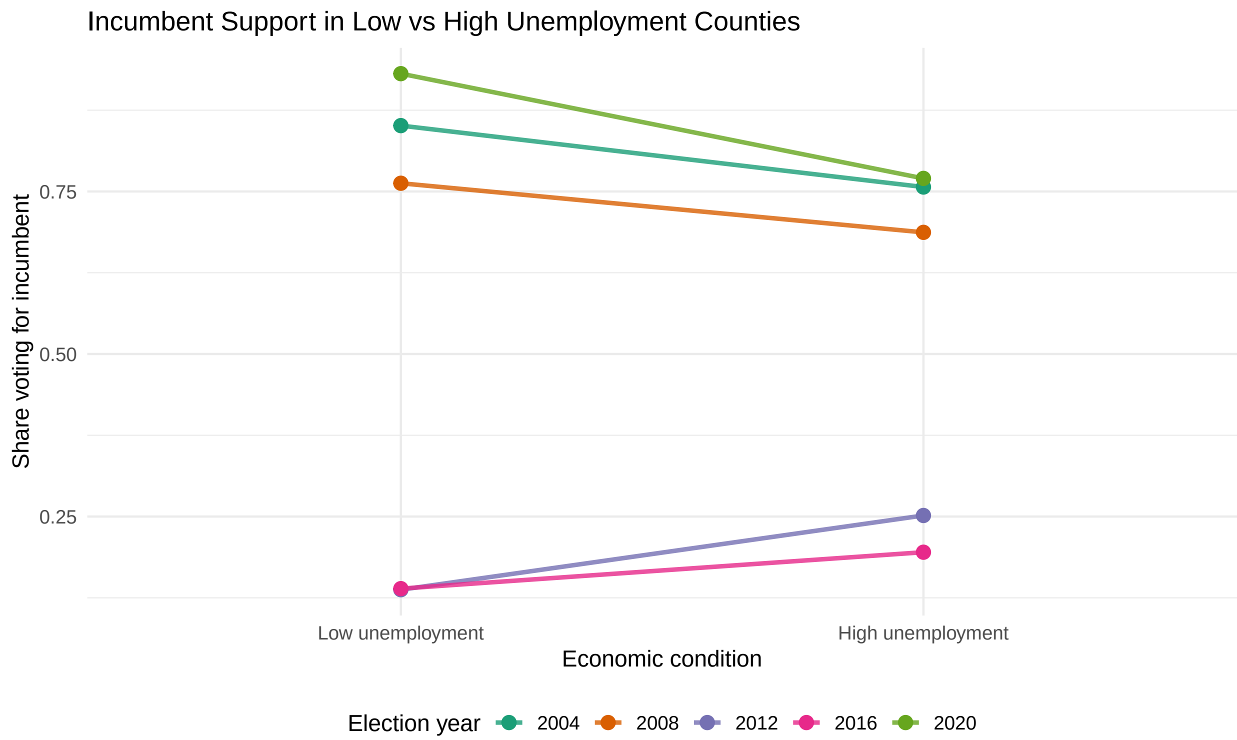
```
                            model_rq2
Dependent Var.:         incumbent_win

Unemployment rate -0.0964*** (0.0217)
Fixed-Effects:    -------------------
FIPS                              Yes
YEAR                              Yes
_________________ ___________________
S.E. type                         IID
Observations                   15,417
Squared Cor.                  0.43868
Pseudo R2                     0.32435
BIC                          44,309.4
---
Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

C. Additional Figures

C.1 Turnout by Unemployment Tier and County Type



C.2 Incumbent Support in Low vs High Unemployment Counties



D. Limitations and Diagnostics (Brief)

- Residual checks for OLS turnout models show no major non-linear patterns, though heteroskedasticity motivates using fixed effects and robust inference.
- Convergence for the FE logit model is satisfactory; results are stable across reasonable model variants.
- Robustness checks (not fully shown here) include alternative county-type classifications and dropping extreme values of turnout and unemployment.

E. Reproducibility Notes

- All analysis relies on the cleaned merged dataset:
  `../data/processed/merged_voting_unemployment_complete.csv`
- Full preprocessing for the unemployment series and voting population is documented in your data preparation scripts and notebooks.
- Random seeds are not required for the core models, as there is no simulation-based component in the final specifications.